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Final Year Project Report

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Theme

Collaborative Question-Answering Platform
Targeted by Academic Specialty

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Dedication

Acknowledgements

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General introduction

Over the past few decades, computer science has emerged as one of the most transformative disciplines of the modern era. From the development of the internet to the rise of artificial intelligence, computing technologies have fundamentally reshaped the way human beings communicate, collaborate, and acquire knowledge. Among the most significant contributions of computer science to society is its capacity to bridge distances and break down barriers — enabling individuals separated by geography, language, and background to exchange ideas and solve complex problems together in real time. This power of connectivity, rooted in software systems and network architectures, continues to redefine the boundaries of what is possible in every domain of human activity, including education. In the academic sphere, the digital revolution has introduced new modalities of learning and knowledge exchange that extend well beyond the traditional classroom. Universities and higher education institutions worldwide are increasingly adopting technology-driven solutions to support their communities, facilitate communication between students and faculty, and promote collaborative learning environments. Yet, despite the widespread availability of general-purpose digital tools, many academic communities still struggle to find platforms that are specifically tailored to their needs — platforms that reflect the structure of academic institutions, the diversity of disciplines, and the nature of scholarly inquiry. In this context, academic communication remains a persistent challenge. Students frequently encounter difficulties in obtaining timely, reliable answers to their course-related questions, while professors face limitations in efficiently disseminating knowledge beyond scheduled class sessions. Existing general-purpose platforms — such as public forums and social media groups — fail to provide the structured, domain-specific environment that academic interaction demands. These solutions often lack mechanisms for quality control, subject-based organization, and institutional alignment, resulting in fragmented and unreliable information exchanges that hinder effective learning. It is in response to these shortcomings that the present project was conceived. This work proposes the design and development of a dedicated collaborative platform aimed at improving academic communication and knowledge sharing between students and professors. At the core of the platform lies a structured question-and-answer system that allows users to post queries, provide answers, and engage in constructive academic discourse within a clearly organized digital environment. To ensure the reliability and quality of contributions, the platform incorporates a voting mechanism through which users can evaluate and rank responses, thereby promoting the most accurate and helpful content to the forefront. Furthermore, the platform is designed around a hierarchical content organization model, whereby academic content is grouped into dedicated spaces according to universities, fields of study, and individual subjects. This structure ensures that all contributions remain contextually relevant and easily accessible, enabling users to navigate directly to the information that pertains to their specific academic context. The result is a focused, noise-free environment that stands in sharp contrast to the disorganized nature of generic online discussion platforms. The overarching objective of this project is to provide a simple, accessible, and effective solution that supports peer-to-peer learning, encourages academic collaboration, and addresses the well-documented limitations of existing tools. By combining intuitive design with purposeful functionality, the proposed platform seeks to create a meaningful and lasting impact on the academic experience of its users. The remainder of this report is organized as follows. The subsequent chapters present a thorough review of existing related works and solutions, followed by a detailed exposition of the system requirements, architectural design, and implementation choices. Finally, the report concludes with an evaluation of the platform's performance and a discussion of potential avenues for future development and improvement.

Chapter 1

General Overview

Introduction

The rapid development of information and communication technologies has profoundly transformed many aspects of modern life, and education is no exception. The emergence of the Internet and digital tools has opened new possibilities for knowledge sharing, giving rise to innovative learning approaches that go beyond the walls of the traditional classroom. In this context, e-learning has established itself as a key component of modern education, offering flexible, accessible, and collaborative learning experiences. This chapter provides a general overview of the e-learning concept and its role in shaping the future of academic education.

0.1 E-Learning and Digital Education

The rapid expansion of digital technologies has deeply transformed the way knowledge is transmitted and acquired. Among the most significant outcomes of this transformation is the emergence of **e-learning** — a mode of education that leverages digital tools and the Internet to deliver learning experiences beyond the boundaries of the traditional classroom. This chapter defines e-learning, examines its main types, highlights its benefits in academic environments, and discusses its connection to our project.

0.2 Definition of E-Learning

E-learning (electronic learning) refers to the use of digital technologies — including computers, smartphones, tablets, and the Internet — to deliver, support, and enhance teaching and learning. It encompasses a broad spectrum of formats: online courses, virtual classrooms, educational videos, interactive exercises, digital collaboration tools, and discussion forums.

Unlike traditional education, which is confined to a physical space and a fixed schedule, e-learning offers learners the flexibility to access content at any time and from any location. As William Horton states in *E-Learning by Design* (2012):

> Effective e-learning is not simply about moving traditional content online, but about fundamentally rethinking how knowledge is structured, delivered, and assessed in a digital environment.

This distinction is crucial: e-learning is not a mere digitization of existing materials, but a redesign of the educational experience itself.

0.3 Types of E-Learning

E-learning can take several forms depending on the level of interactivity and the timing of communication between learners and instructors.

0.3.1 Synchronous E-Learning

Synchronous e-learning takes place in real time, with students and instructors interacting simultaneously over a digital medium. Examples include live video lectures, webinars, and virtual classrooms. This format closely mirrors the dynamics of a physical classroom while removing geographical constraints.

0.3.2 Asynchronous E-Learning

Asynchronous e-learning is self-paced: learners access pre-recorded content, downloadable materials, or discussion forums at a time of their choosing. There is no requirement to be online simultaneously with others. This is the most flexible form of e-learning and is widely used in online courses and academic platforms.

0.3.3 Blended Learning

Blended learning (also called hybrid learning) combines face-to-face instruction with digital tools and online activities. It preserves the benefits of direct human interaction while integrating the flexibility and accessibility of digital resources.

0.4 Examples of E-Learning Platforms

Several e-learning platforms have been developed specifically for academic environments. Among the most notable are the following:

0.4.1 Moodle

(Modular Object-Oriented Dynamic Learning Environment) is an open-source Learning Management System (LMS) widely adopted by universities around the world, including Algerian universities. It allows instructors to post course materials, manage assignments, and interact with students through forums and quizzes.

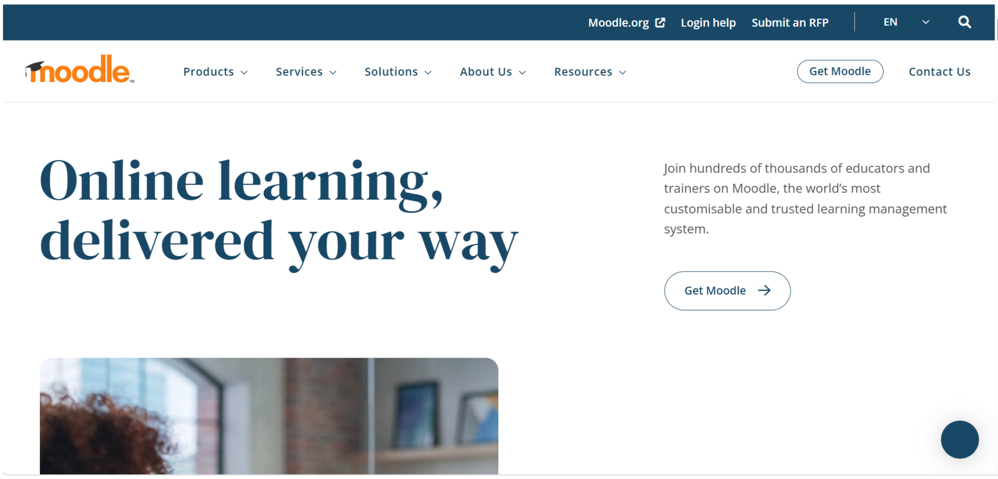

Figure 1: moodle platform interface

0.4.2 Google Classroom

is a free platform developed by Google, designed to simplify the management of courses and assignments. It provides a straightforward environment for communication between students and teachers, making it one of the most accessible tools in academic institutions.

Figure 2: google classroom platform interface

0.4.3 Piazza

is a collaborative Q&A platform designed specifically for university courses. It allows students to post questions related to their courses and receive answers from both their peers and instructors, encouraging a culture of collaborative learning.

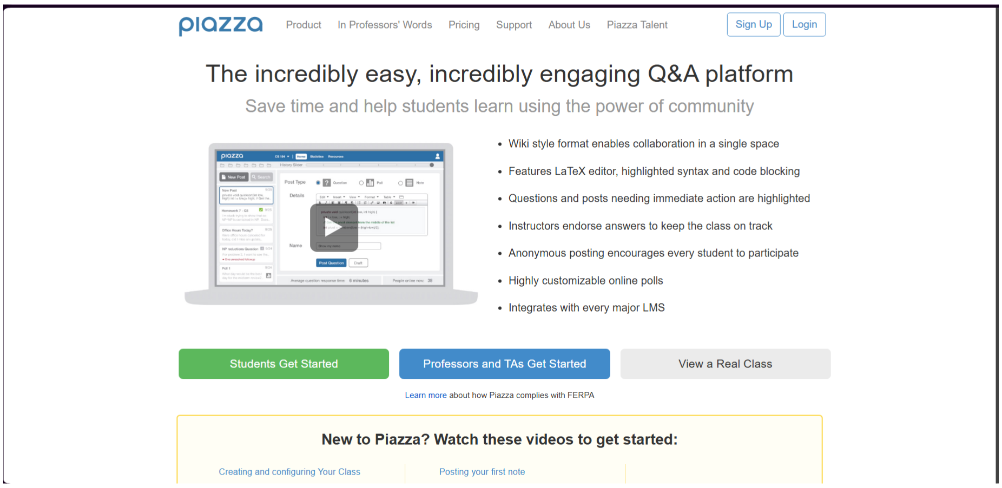

Figure 3: piazza platform interface

0.5 Benefits of E-Learning in Academic Environments

E-learning offers numerous advantages that make it particularly well-suited to academic settings:

- **Accessibility:** Students can access course materials, resources, and support at any time and from any location, removing barriers related to physical distance or scheduling conflicts.

- **Flexibility:** Learners can progress at their own pace, revisit difficult concepts, and organize their study time independently — which is especially valuable in higher education where students manage multiple subjects simultaneously.

- **Collaboration:** Digital platforms enable students to interact with peers and instructors beyond classroom hours — through forums, messaging systems, shared documents, and Q&A tools — fostering a culture of collaborative learning.

- **Scalability:** A single e-learning platform can serve hundreds or thousands of students at once, making knowledge accessible at a scale that traditional classrooms cannot achieve.

- **Engagement:** Multimedia content, interactive exercises, and gamification elements can increase student motivation and engagement compared to passive, lecture-based learning.

- **Traceability:** Digital platforms allow instructors to track student activity, monitor progress, and identify learners who may need additional support.

0.6 E-Learning and Collaborative Knowledge Sharing

One of the most powerful dimensions of e-learning is its capacity to support **collaborative knowledge sharing**. Rather than treating students as passive recipients of information, modern e-learning platforms position them as active contributors to a shared learning community.

This idea is rooted in the constructivist theory of learning, which argues that knowledge is not simply transmitted from teacher to student, but actively constructed through interaction, discussion, and problem-solving. Digital environments are particularly well-suited to this approach: they provide spaces where students can ask questions, offer answers, debate ideas, and build upon each other's contributions.

This principle is at the heart of our project. By combining e-learning with a structured Q&A mechanism, our platform transforms students from passive consumers of academic content into active participants in the production of knowledge within their specific field of study.

0.7 Conclusion

E-learning has evolved from a supplementary tool into a fundamental pillar of modern education. Its flexibility, accessibility, and capacity for collaboration make it an ideal foundation for academic digital platforms. The limitations of existing general-purpose platforms — particularly their lack of specialization by academic field — reveal a clear need for a more targeted solution. This need motivates the design of our **Collaborative Question-Answering Platform Targeted by Academic Specialty**, which will be presented in the following chapter.

Chapter 2

Description of UML Language and Development Tools

0.8 Introduction

As software systems grow increasingly complex, new methods, languages, and tools have been developed to better understand client needs and requirements, with the goal of building systems that truly meet expectations. Among these approaches, Object-Oriented Programming (OOP) stands out as one of the most widely adopted, and it is within this paradigm that the UML modeling language finds its place. The development of any web-based system must be preceded by a structured analysis and design methodology. This methodology serves as a roadmap for formalizing the preliminary stages of development, ensuring that the final product remains faithful to the client's needs and expectations.

0.9 UML Modeling

0.9.1 Use Case Diagram

0.9.2 Sequence Diagram

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0.9.3 Class Diagram

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0.10 Database Design

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0.11 Conclusion

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Chapter 1

Implementation

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1.2 Technologies Used

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1.2.2 Backend

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1.2.3 Database

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1.3 Application Interfaces

1.3.1 Authentication Page

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1.3.2 Home Page

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1.4 Conclusion

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Chapter 2

General Conclusion

This project presented the design and implementation of a collaborative Q&A web platform targeted by academic specialty. Through this work, we gained practical experience in web development, database design, and system modeling using UML...

As future perspectives, we envision adding AI-based question recommendation, mobile application support, and integration with university course management systems.